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How Hall electric fields intrinsically chaotize and heat ions during collisionless magnetic reconnection

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The origin of extreme and anomalous ion heating during magnetic reconnection has been a longstanding mystery. It is shown in this paper that, at the core of the process, the stochastic heating mechanism is responsible for this energization. The in-plane Hall electric fields that intrinsically develop during collisionless reconnection not only accelerate ions, but also thermalize them by inducing chaotic motion. The mechanism explains various observed features of ion heating, e.g., its relation to the guide field strength and the ion mass-to-charge ratio.

I. INTRODUCTION

Magnetic reconnection is a plasma process through which magnetic energy is converted into kinetic energy via magnetic topology changes[1]. Although magnetic reconnection is forbidden in ideal magneto-hydrodynamics (MHD) due to the connected property of magnetic field lines, it is now widely accepted that collisionless effects (e.g., the Hall effect) that become important at kinetic scales break this conservation law and allow for fast reconnection rates consistent with observations[2]. Despite decades of research, however, mysteries remain regarding the exact mechanism behind the magnetic to kinetic energy conversion, which is fundamentally what magnetic reconnection is.

A phenomenon that has been particularly puzzling is the accompaniment of magnetic reconnection by anomalous ion heating that is much faster and stronger than heating by conventional mechanisms. This feature has been distinguished in myriads of astrophysical and laboratory reconnection events[3–12] and typically exhibits the following characteristics: (i) association with in-plane Hall electric fields[8,10,13,14], (ii) non-conservation of the ion magnetic moment μ[15–17], (iii) temperature anisotropy ($T_{i\perp} \neq T_{i\parallel}$)[16,18–20], (iv) reduction of the heating amount and extent with the increase of the guide field (background out-of-plane magnetic field)[16], and (v) preferential heating of ions with higher mass-to-charge ratios m_i/q_i[21–23].

Because anomalous ion heating is an important element of other critical, longstanding problems such as the coronal heating problem[24], various mechanisms have been proposed to resolve this conundrum. These include turbulent/multiple-island interactions[25–29], pickup[15,16,30], remagnetization[10], and shock/wave interactions[9,23], but there has been little to no consensus on a comprehensive mechanism.

Since fast ion heating associated with magnetic reconnection is observed across an extremely wide range of plasma parameters, one can expect there to be a mechanism that is intrinsic to the reconnection process itself. In this paper, it will be shown that the in-plane Hall electric fields that intrinsically arise during collisionless magnetic reconnection render ion motions chaotic and *de facto* heat the species through phase-space broadening. This paper is organized as follows. In Sec. II, the so-called "stochastic heating" mechanism and the criterion for its occurrence will be illustrated in two different theoretical contexts: (i) an electrostatic wave and (ii) an inverse parabolic potential. In Sec. III, the criterion for stochastic ion heating in a generic collisionless reconnection geometry will be derived in a fluid context. In Sec. IV, the condition for chaotic ion motion in the kinetic Harris equilibrium will be derived. In Sec. V, the mechanism will be verified through results from particle-in-cell simulations. Finally, each of the aforementioned features of ion heating during reconnection will be individually examined in relation to the stochastic heating mechanism in Sec. VI. All in all, this paper aims to summarize and expand on previous relevant studies[31,32] and the corresponding invited talk at the American Physical Society, Division of Plasma Physics meeting in Memphis, Tennessee in 2020.

II. ORBIT CHAOTIZATION AND STOCHASTIC HEATING

Stochastic heating is a mechanism in which strong electric fields perpendicular to the local magnetic field chaotize particle orbits. If the electric field acts in a way that energizes the particles, then an effective heating occurs due to broadening in phase space. The condition for stochastic heating is[33–35]

$$\alpha\left(\mathbf{x},t\right)=\frac{m_\sigma}{q_\sigma\left|\mathbf{B}\right|^2}\left|\nabla_\perp\cdot\mathbf{E}_\perp\right|>1, \tag{1}$$

where m_σ and q_σ are respectively the mass and charge of species σ, $\mathbf{B}=\mathbf{B}\left(\mathbf{x},t\right)$ is the local magnetic field, $\mathbf{E}_\perp=\mathbf{E}_\perp\left(\mathbf{x},t\right)$ is the electric field perpendicular to $\mathbf{B}$, and $\nabla_\perp$ is the component of the gradient operator in the direction perpendicular to $\mathbf{B}$. α is the parameter that yields a spatio-temporal prediction for stochastic heating.

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A. Electrostatic Wave

The origin of Eq. 1 can be understood by examining the motion of a particle in an electrostatic wave[36,37]. Consider a particle subject to the following electric and magnetic fields:

$$\mathbf{E} = \hat{x} E_0 \sin(kx - \omega t), \tag{2}$$

$$\mathbf{B} = \hat{z} B_0. \tag{3}$$

The first-order motion of the particle is the $\mathbf{E} \times \mathbf{B}$ drift, which is in the y-direction:

$$\mathbf{v}_E = -\hat{y} \frac{E_0}{B_0} \sin(kx - \omega t). \tag{4}$$

Since the magnetic field is uniform, the next-order motion is the polarization drift:

$$\mathbf{v}_p = \frac{m_\sigma}{q_\sigma B_0^2} \frac{d\mathbf{E}}{dt}, \tag{5}$$

$$= \hat{x} \frac{m_\sigma E_0}{q_\sigma B_0^2} \frac{d}{dt} \sin(kx - \omega t), \tag{6}$$

$$= \hat{x} \frac{m_\sigma E_0}{q_\sigma B_0^2} \left(k \frac{dx}{dt} - \omega \right) \cos(kx - \omega t). \tag{7}$$

Here, because $\mathbf{v}_E$ is in the y-direction, $\hat{x} dx/dt = \mathbf{v}_p$. Solving for $\mathbf{v}_p$,

$$\mathbf{v}_p = -\hat{x} \frac{\omega}{k} \frac{\alpha \cos(kx - \omega t)}{1 - \alpha \cos(kx - \omega t)}, \tag{8}$$

where

$$\alpha = \frac{m_\sigma k E_0}{q_\sigma B_0^2}, \tag{9}$$

which is the stochastic heating parameter in the present case. It can be seen that when $\alpha > 1$, $\mathbf{v}_p$ is divergent. This means that the guiding center approximation breaks down, and thus cyclotron motion is destroyed, even for wave frequencies $\omega \ll \omega_c = q_\sigma B_0 / m_\sigma$. The particle undergoes stochastic motion in phase space and is effectively heated in the perpendicular direction, and evidence for this heating has been observed in laboratory experiments[35,38] and spacecraft measurements[36].

If two particles that initially had an infinitesimal separation in phase space diverge from each other in time, the system is extremely sensitive to the initial conditions. This is the concept of Lyapunov exponents[39], and a positive exponent indicates a chaotic system. When Eq. 1 is satisfied, the system has a positive Lyapunov exponent[36]. To see why this is so, let us consider two particles whose coordinates in position/velocity space are respectively $(\mathbf{x}_1, \dot{\mathbf{x}}_1)$ and $(\mathbf{x}_2, \dot{\mathbf{x}}_2)$. Denoting their separations $(\delta\mathbf{x}, \delta\dot{\mathbf{x}})$, the perpendicular equations of motion are, to first order in δx,

$$\delta\ddot{x} = \frac{q_\sigma}{m_\sigma} \frac{\partial E_x}{\partial x} \delta x + \frac{q_\sigma B_0}{m_\sigma} \delta\dot{y}, \tag{10}$$

$$\delta\ddot{y} = -\frac{q_\sigma B_0}{m_\sigma} \delta\dot{x}. \tag{11}$$

Figure 1. Trajectories of five particles subject to an electrostatic wave with (a) $\alpha = 0.5$ and (b) $\alpha = 1.5$. The particles are randomly chosen from an ensemble of 1,000 particles. Note the different axis scales of (a) and (b). Logarithms of the average kinetic energy and thermal energy of the ensemble as a function of time for (c) $\alpha = 0.5$ and (d) $\alpha = 1.5$.

Taking the time derivative of Eq. 10 and substituting in Eq. 11,

$$\delta\dddot{x} = \frac{q_\sigma^2 B_0^2}{m_\sigma^2} \left(\frac{m_\sigma}{q_\sigma B_0^2} \frac{\partial E_x}{\partial x} - 1 \right) \delta\dot{x}, \tag{12}$$

$$= \frac{q_\sigma^2 B_0^2}{m_\sigma^2} (\alpha - 1) \delta\dot{x}. \tag{13}$$

It can clearly be seen that Eq. 13 yields an exponential growing solution for $\delta\dot{x}$ when $\alpha > 1$. Therefore, stochastic heating involves chaotic particle motion.

In order to verify stochastic heating by an electrostatic wave, 1,000 particle trajectories were integrated under the fields defined by Eqs. 2 and 3 for $\alpha = 0.5$ and 1.5, the latter of which should trigger stochastic heating. The non-relativistic Boris algorithm[40,41] was used; length and time were normalized to k^{-1} and ω_c, respectively. The wave frequency was set as $\omega/\omega_c = 0.4$, and the particles were initialized at $\mathbf{x} = 0$ with small random velocities corresponding to a thermal energy of $E/m_\sigma k^{-2} \omega_c^2 = 10^{-6}$.

Figure 1(a) shows 5 randomly chosen particle trajectories for $\alpha = 0.5$. The trajectories do not differ much from one another, indicating predictable motion. This is in contrast to the 5 trajectories in Fig. 1(b) which clearly exhibit unpredictability and chaos for $\alpha = 1.5 > 1$ (also note that the axis scales are different from those in Fig. 1(a)). Figure 1(c) shows the time-dependent kinetic and thermal energies of the particle ensemble for $\alpha = 0.5$. These correspond to the mean and the standard deviation of the particle energy distribution. Both energies initially increase due to the particles' starting position in the wave potential, but subsequently undergo oscillations about their mean values which stay constant. For $\alpha = 1.5$ (Fig. 1(d)), however, there is a clear increase of thermal energy—i.e., stochastic heating—until a stagnation point. The increase is linear in log scale and so is exponential in linear scale. Figure 1(d) shows that the temperature has increased by nearly three orders of magnitude.

Although α was introduced as a spatio-temporal predictor of heating, one must be careful not to naively relate the value of α to the heating amount. This is because, while Eq. 1 applies generally at frequencies comparable to or less than several times the species' cyclotron frequency $f_{c\sigma}$, in the opposite case where $f \gg f_{c\sigma}$, an additional condition must be satisfied, namely that the species' thermal velocity must be comparable to the wave phase velocity[42–44]. If this additional condition is not satisfied, then the species is unaffected by waves with frequencies $f \gg f_{c\sigma}$. In this paper, only the former case

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Figure 2. Trajectories of 100 particles subject to an inverse parabolic potential with (a) $\alpha = 0.4$ and (b) $\alpha = 1.04$. The particles are randomly chosen from an ensemble of 10,000 particles. Note the different axis scales of (a) and (b). Logarithms of the average kinetic energy and thermal energy of the ensemble as a function of time for (c) $\alpha = 0.4$ and (d) $\alpha = 1.04$.

where $f \lesssim f_{c\sigma}$ will be considered.

B. Inverse Parabolic Potential

Stochastic heating can also occur in static electric fields[36,45,46]. Let us consider a parabolic electrostatic potential in cylindrical geometry, or

$$\phi = -\frac{1}{2}\frac{E_0}{\lambda_r}r^2, \tag{14}$$

which generates a corresponding electric field

$$\mathbf{E} = E_0\frac{\mathbf{r}}{\lambda_r}. \tag{15}$$

Assuming a uniform magnetic field of $\mathbf{B} = B_0\hat{z}$, the perpendicular equations of motion are

$$\ddot{x} = \frac{q_\sigma}{m_\sigma}\left(E_0\frac{x}{\lambda_r} + B_0\dot{y}\right), \tag{16}$$

$$\ddot{y} = \frac{q_\sigma}{m_\sigma}\left(E_0\frac{y}{\lambda_r} - B_0\dot{x}\right). \tag{17}$$

Defining $\tilde{x} = x + iy$, $\omega_c = q_\sigma B_0/m_\sigma$, and $\alpha = 4E_0/\lambda_r B_0\omega_c$, we have

$$\ddot{\tilde{x}} = \omega_c^2\alpha\tilde{x} - i\omega_c\dot{\tilde{x}}. \tag{18}$$

Note that α defined here differs from that in Eq. 1 by a numerical factor of 2. Assuming a dependence $\tilde{x} \sim e^{i\omega t}$,

$$\omega^2 + \omega_c\omega + \omega_c^2\alpha = 0, \tag{19}$$

for which the solution is

$$\frac{\omega}{\omega_c} = \frac{-1 \pm \sqrt{1-\alpha}}{2}. \tag{20}$$

It can be seen that when $\alpha > 1$, Eq. 20 yields an exponentially growing solution for $\tilde{x}$.

The particle simulation in Sec. II A was slightly altered to verify chaotic motion in an inverse parabolic potential. 10,000 particles were initialized at $\mathbf{x} = 0.01\lambda_r$ with small random velocities corresponding to a thermal energy of $E/m_\sigma\lambda_r^2\omega_c^2 = 10^{-6}$. Figure 2(a) shows 100 randomly chosen particle trajectories for $\alpha = 0.4$. The particles undergo predictable $\mathbf{E}\times\mathbf{B}$ motion in the azimuthal direction, and their kinetic and thermal energies oscillate about the initial values, as shown in Fig. 2(c). In Fig. 2(b), however, the absolute positions of the particles and their separations from one another diverge for $\alpha = 1.04 > 1$ (note the larger axis scales than those in Fig. 2(a)). This divergence leads to an exponential increase in both the kinetic and thermal energies of the ensemble, as shown in Fig. 2(d). Thus, the energy garnered from falling down a potential hill is converted into both the average kinetic energy and thermal energy of the ensemble, the latter of which corresponds to stochastic heating.

III. GENERIC COLLISIONLESS RECONNECTION GEOMETRY

A. Connected Nature of Canonical Vorticity

At this point, let us segue into a fundamental physical property, namely the connected nature of canonical vorticity. Consider the electron fluid equation of motion without the pressure term,

$$m_e\left(\frac{\partial\mathbf{u}_e}{\partial t} + \mathbf{u}_e\cdot\nabla\mathbf{u}_e\right) = q_e\left(\mathbf{E} + \mathbf{u}_e\times\mathbf{B}\right), \tag{21}$$

where m_e and q_e are respectively electron mass and charge, and $\mathbf{u}_e$ is the electron fluid velocity. Taking the curl of Eq. 21, using the vector calculus identity $\mathbf{u}_e\cdot\nabla\mathbf{u}_e = \nabla\left(u_e^2/2\right) - \mathbf{u}_e\times(\nabla\times\mathbf{u}_e)$, and using Faraday's law $\nabla\times\mathbf{E} = -\partial\mathbf{B}/\partial t$, we have

$$\frac{\partial\mathbf{Q}_e}{\partial t} = \nabla\times\left(\mathbf{u}_e\times\mathbf{Q}_e\right), \tag{22}$$

where

$$\mathbf{Q}_e = m_e\nabla\times\mathbf{u}_e + q_e\mathbf{B} \tag{23}$$

is the electron "canonical vorticity," which is the curl of the electron canonical momentum.

The plasma induction equation in ideal MHD,

$$\frac{\partial\mathbf{B}}{\partial t} = \nabla\times\left(\mathbf{U}\times\mathbf{B}\right), \tag{24}$$

asserts that magnetic flux is conserved in the plasma frame and that magnetic field lines cannot reconnect. Because Eq. 22 has the same mathematical form as Eq. 24, it asserts that the flux associated with $\mathbf{Q}_e$ is conserved and that $\mathbf{Q}_e$ field lines cannot reconnect. However, because $\mathbf{Q}_e$ is a combination of electron vorticity and magnetic field, the latter is free to convert into the former while preserving $\mathbf{Q}_e$ flux; i.e., reconnection of $\mathbf{B}$ is allowed while that of $\mathbf{Q}_e$ is not.

As a sidenote, the pressure term which in principle should be present in Eq. 21 may be dropped by either positing low electron plasma beta β_e or barotropic pressure, i.e., $\nabla p_e \parallel \nabla n_e$ where p_e and n_e are respectively electron pressure and density[47]. Effects due to the full pressure tensor are encapsulated in the "canonical battery" term[48].

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Figure 3. A generic collisionless magnetic reconnection geometry. Magnetic field lines (white lines) reconnect while canonical vorticity field lines (red lines) do not and pile up near $x = 0$. The color qualitatively shows the potential that generates the in-plane Hall electric fields. © AAS. Adapted with permission from Astrophys. J. Lett. 868(2), L31 (2019)[31].

B. Calculation of the Electric Fields

Now let us calculate the electric fields in a generic collisionless reconnection geometry and see whether or not they satisfy the stochastic ion heating criterion. Figure 3 shows an example of such a geometry. The sheared field lines point in the $\pm y$-directions, while reconnection corresponds to having a finite x-component. The system is two-dimensional, i.e., $\partial/\partial z \to 0$, and the initial out-of-plane magnetic field is zero. It can be seen that $\mathbf{B}$ field lines (white lines) reconnect while $\mathbf{Q}_e$ field lines (red lines) do not and instead pile up near $x = 0$. The color represents the potential generating the Hall electric fields.

The expression for the electric field may be obtained by rearranging the electron equation of motion:

$$\mathbf{E} = \frac{m_e}{q_e}\left(\frac{\partial \mathbf{u}_e}{\partial t} + \mathbf{u}_e \cdot \nabla \mathbf{u}_e\right) - \mathbf{u}_e \times \mathbf{B}. \tag{25}$$

In terms of the canonical vorticity $\mathbf{Q}_e$, Eq. 25 becomes

$$\mathbf{E} = \frac{m_e}{q_e}\left(\frac{\partial \mathbf{u}_e}{\partial t} + \nabla\left[\frac{u_e^2}{2}\right]\right) - \frac{1}{q_e}\mathbf{u}_e \times \mathbf{Q}_e. \tag{26}$$

Justifications for dropping the pressure term are described in Yoon and Bellan[31].

The time-dependent term $\partial \mathbf{u}_e/\partial t$ in Eq. 26 may be rewritten in terms of spatial gradients by solving for the growth rate of the system. Consider the y-component of Eq. 22:

$$\frac{\partial Q_{ey}}{\partial t} = \mathbf{Q}_e \cdot \nabla u_{ey} - \mathbf{u}_e \cdot \nabla Q_{ey} - Q_{ey}\left(\nabla \cdot \mathbf{u}_e\right). \tag{27}$$

Near $x = 0$, $Q_{ex} = 0$ because $\mathbf{Q}_e$ cannot reconnect as can be seen by the red lines in Fig. 3. Assuming incompressibility ($\nabla \cdot \mathbf{u}_e$) due to quasi-neutrality, we thus have

$$\left(\frac{\partial}{\partial t} + \mathbf{u}_e \cdot \nabla\right) Q_{ey} = \frac{\partial u_{ey}}{\partial y} Q_{ey}, \tag{28}$$

$$\frac{1}{Q_{ey}}\frac{DQ_{ey}}{Dt} = \frac{\partial u_{ey}}{\partial y}. \tag{29}$$

where $D/Dt = \partial/\partial t + \mathbf{u}_e \cdot \nabla$. The growth rate of the system can thus be expressed as

$$\frac{\partial}{\partial t} \simeq \frac{\partial u_{ey}}{\partial y} \simeq -\frac{\partial u_{ex}}{\partial x}. \tag{30}$$

Let us now first analyze Eq. 26 along the inflow direction, i.e., the black dashed line in Fig. 3. The x-component of Eq. 26 is

$$E_x = \frac{m_e}{q_e}\left(\frac{\partial u_{ex}}{\partial t} + \frac{\partial}{\partial x}\left[\frac{u_e^2}{2}\right]\right) - \frac{1}{q_e}\left(u_{ey}Q_{ez} - u_{ez}Q_{ey}\right). \tag{31}$$

Noting that the outflow $u_{ey} = 0$ along the black dashed line due to symmetry and using Eq. 30,

$$E_x \simeq \frac{m_e}{q_e}\left(-u_{ex}\frac{\partial u_{ex}}{\partial x} + \frac{\partial}{\partial x}\left[\frac{u_{ex}^2 + u_{ez}^2}{2}\right]\right) + \frac{1}{q_e}u_{ez}Q_{ey}, \tag{32}$$

$$= \frac{m_e}{q_e}\left(\frac{\partial}{\partial x}\left[\frac{u_{ez}^2}{2}\right]\right) + \frac{1}{q_e}u_{ez}\left(-m_e\frac{\partial u_{ez}}{\partial x} + q_e B_y\right), \tag{33}$$

$$= u_{ez}B_y. \tag{34}$$

Using Ampère's law $\nabla \times \mathbf{B} \simeq -\mu_0 n_e e \mathbf{u}_e$,

$$E_x \simeq -\frac{B_y}{\mu_0 n_e e}\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right) \tag{35}$$

$$\simeq -\frac{B_y}{\mu_0 n_e e}\left(\frac{\partial B_y}{\partial x}\right) \tag{36}$$

$$= -\frac{\partial}{\partial x}\left(\frac{B_y^2}{2\mu_0 n_e e}\right) \tag{37}$$

where the fact that $B_x = 0$ along the inflow was used in the second line. The electric field along the inflow direction has therefore been expressed in the form of an effective potential, and this electric field may be inserted into Eq. 1 to yield the condition for stochastic ion heating in the inflow direction[31]:

$$L_x^2 < d_i^2, \tag{38}$$

where $L_x = \left|B_y^{-1}\partial B_y/\partial x\right|^{-1}$ is the magnetic field shear length scale, and $d_i = c/\sqrt{ne^2/m_i\epsilon_0}$ is the ion skin depth. Interestingly, Eq. 38 is the condition for collisionless magnetic reconnection to occur and so is intrinsically satisfied.

Let us now examine the electric field in the outflow direction, i.e., the magenta dashed line in Fig. 3. The y-component of Eq. 26 is

$$E_y = \frac{m_e}{q_e}\left(\frac{\partial u_{ey}}{\partial t} + \frac{\partial}{\partial y}\left[\frac{u_e^2}{2}\right]\right) - \frac{1}{q_e}\left(u_{ez}Q_{ex} - u_{ex}Q_{ez}\right). \tag{39}$$

Noting that, along the magenta line, $Q_{ex} = 0$ due to non-reconnection of $\mathbf{Q}_e$ and $u_{ex} = 0$ due to symmetry and using Eq. 30,

$$E_y = -\frac{\partial}{\partial y}\left(\frac{m_e}{e}\frac{u_{ey}^2 + u_e^2}{2}\right). \tag{40}$$

Inserting this into Eq. 1 yields the condition for stochastic ion heating in the outflow direction[31],

$$\frac{m_i}{m_e} > 1, \tag{41}$$

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Figure 4. A graphical representation of the Harris equilibrium. A plasma sheath (color; yellow signifies high density) is confined by sheared magnetic fields (white lines). An electric field (green arrows) develops in the center-of-mass frame.

which is always satisfied.

All in all, it has been shown using the connected property of canonical vorticity field lines that the stochastic ion heating condition is always satisfied in both the inflow and outflow directions in a generic collisionless magnetic reconnection geometry. Therefore, stochastic ion heating is intrinsic to the process.

IV. KINETIC HARRIS EQUILIBRIUM

An important system that has profound connections to magnetic reconnection is the Harris equililbrium[49], which describes a fully kinetic stationary one-dimensional solution to the Vlasov equation for a plasma sheath confined by opposing magnetic fields. The system is solved in a frame where ions and electrons have equal and opposite average velocities in the $\hat{z}$-direction, i.e., $\mathbf{V}_i = V\hat{z} = -\mathbf{V}_e$. The solutions for the magnetic field, sheath density, and the electric field in this frame are

$$\mathbf{B}' = \hat{y}2\sqrt{\mu_0 n_0 k_B T}\tanh\frac{x}{\lambda}, \tag{42}$$

$$n' = n_0\cosh^{-2}\frac{x}{\lambda}, \tag{43}$$

$$\mathbf{E}' = 0, \tag{44}$$

where n_0 and T are the sheath peak density and temperature, respectively. λ is equilibrium sheath half-thickness which is equal to $\lambda = \lambda_D c/V$ where $\lambda_D = \sqrt{\epsilon_0 k_B T/n_0 e^2}$. The prime indicates that the solutions are NOT solved in a center-of-mass frame.

Lorentz transforming to the stationary-ion frame, which is approximately the center-of-mass frame due to the heavy ion mass, we have

$$\mathbf{B} = \mathbf{B}' = \hat{y}2\sqrt{\mu_0 n_0 k_B T}\tanh\frac{x}{\lambda}, \tag{45}$$

$$\mathbf{E} = V\hat{z}\times\mathbf{B}' = -\hat{x}2V\sqrt{\mu_0 n_0 k_B T}\tanh\frac{x}{\lambda}. \tag{46}$$

Here we have assumed $V \ll c$ and so the transformation gamma $\gamma = 1$. Figure 4 graphically illustrates the equilibrium. The plasma sheath (color) is confined by opposing magnetic fields (white lines), and electric fields are present (green arrows) which point toward the center of the sheath.

Inserting Eqs. 45 and 46 into Eq. 1 while including a background out-of-plane magnetic field (guide field) $B_z\hat{z} = \hat{z}b_g B_y|_{x\to\infty}$, the following relation can be obtained[32]:

$$2\frac{\lambda^2}{d_i^2}\cosh^2\left(\frac{x}{\lambda}\right)\left[\tanh^2\left(\frac{x}{\lambda}\right) + b_g^2\right] < 1, \tag{47}$$

Figure 5. (a) $\bar{x}_{\rm sh}/\bar{\lambda}$ as a function of $\bar{\lambda}$ and b_g. (b) $\bar{x}_{\rm sh}/\bar{\lambda}$ as a function of b_g for different values of $\bar{\lambda}$.

Figure 6. Magnetic field lines (black lines) and E_x (color) from particle-in-cell simulations of magnetic reconnection for guide field strengths (a) $b_g = 0$, (b) $b_g = 0.1$, (c) $b_g = 0.3$, and (d) $b_g = 0.5$. © AAS. Reproduced with permission from Astrophys. J. Lett. 887(2), L29 (2019)[32].

where d_i is the ion skin depth. Writing $\bar{x} = x/d_i$ and $\bar{\lambda} = \lambda/d_i$, Eq. 47 can be solved in closed form, yielding the solution $\bar{x} < \bar{x}_{\rm sh}$ where

$$\bar{x}_{\rm sh} = \frac{\bar{\lambda}}{2}\ln\frac{\bar{\lambda}^2\left(1-b_g^2\right)+1+\sqrt{\left(1-2\bar{\lambda}^2 b_g^2\right)\left(2\bar{\lambda}^2+1\right)}}{\bar{\lambda}^2\left(1+b_g^2\right)}. \tag{48}$$

Therefore, $\bar{x}_{\rm sh}$ represents the spatial extent of stochastic ion heating.

A more useful parameter is $\bar{x}_{\rm sh}/\bar{\lambda}$, the fractional extent of stochastic ion heating with respect to the sheath thickness. Figure 5(a) shows $\bar{x}_{\rm sh}/\bar{\lambda}$ as a function of $\bar{\lambda}$ and b_g, and Fig. 5(b) shows $\bar{x}_{\rm sh}/\bar{\lambda}$ as a function of b_g for various values of $\bar{\lambda}$. Four features can be deduced from the solution: (i) stochastic ion heating becomes significant ($\bar{x}_{\rm sh} \gtrsim \bar{\lambda}$) when the sheath thickness is less than or comparable to the ion skin depth, i.e., $\bar{\lambda} \lesssim 1$, (ii) the fractional heating extent increases as sheath thickness decreases, (iii) the fractional heating extent decreases as the guide field increases, and (iv) no stochastic ion heating is predicted above the threshold guide field $b_g = \left(\sqrt{2}\bar{\lambda}\right)^{-1}$ (red dashed line in Fig. 5(a)). The last feature is so because no real solution for $\bar{x}_{\rm sh}$ exists when the argument of the square root in Eq. 48 is negative.

V. PARTICLE-IN-CELL SIMULATIONS

The existence of stochastic ion heating during collisionless magnetic reconnection and the various characteristics predicted in previous sections were verified by results from particle-in-cell (PIC) simulations. The open-source SMILEI code[50] was used, and detailed simulation conditions may be found in Yoon and Bellan[32]. The important simulation parameters here are $\bar{\lambda} = 1$, $m_i/m_e = 100$, and $b_g = 0, 0.1, 0.3, 0.5$.

Figure 6 shows the magnetic field lines (white lines) and E_x (colors) during reconnection for different guide field strengths. It can be seen that although the detailed structure changes as the guide field changes, the electric fields are generally pointing towards $x = 0$. E_y (not plotted) also develops pointing away from the origin, albeit much weaker. The electric field corresponds to that generated by a saddle-like potential like that shown in Fig. 3.

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Figure 7. (a)-(d) Logarithms of the stochastic ion heating criterion α for different guide field strengths b_g. The condition is satisfied in the red and yellow regions. (e)-(h) Ion temperature profiles that are respectively synchronous with (a)-(d). © AAS. Reproduced with permission from Astrophys. J. Lett. 887(2), L29 (2019)[32].

Figures 7(a)-(d) show the logarithm of the stochastic ion heating criterion α for different guide field strengths. The red and yellow colors represent regions where stochastic ion heating is predicted. As anticipated from Sec. IV, the heating extent decreases as the guide field increases, and no stochastic heating is predicted above a threshold guide field ($b_g = 0.5$). The exact value of this threshold is understandably different because Sec. IV examines an equilibrium with a static $\bar{\lambda}$ whereas $\bar{\lambda}$ is dynamic in this situation.

Figures 7(e)-(h) show the ion temperatures for different guide field strengths, respectively synchronous with Figs. 7(a)-(d). It can be seen that the regions of high ion temperatures coincide with regions where stochastic ion heating is predicted. The heating amount and extent decrease with increasing guide field, and extreme ion heating cannot be observed above the threshold guide field. This correlation is a strong evidence that stochastic heating is responsible for the observed anomalous ion heating.

VI. DISCUSSION

In Sec. I, five main features of ion heating during collisionless reconnection were identified. Each of these features will now be individually examined in relation to the stochastic heating mechanism.

1. *The association of anomalous ion heating with Hall electric fields.* It has been shown that Hall electric fields intrinsically satisfy the stochastic ion heating criterion during collisionless magnetic reconnection for up to moderate guide fields ($b_g \sim 0.5$); the electric fields thus render ion motions chaotic and effectively heat the species. In particular, Hall electric fields as well as ion heating were measured *in situ* in the Magnetic Reconnection Experiment (MRX)[10] during a driven collisionless reconnection event. Inserting the measured values of $E \sim 700$ V/m, $B \sim 0.01$ T and a length scale of ~ 1 cm yields $\alpha = 14.6 > 1$, so the fact that Hall electric fields induce stochastic ion heating is well evidenced by experiments.

2. *Violation of the magnetic moment μ conservation law.* Stochastic heating is a mechanism wherein the guiding-center approximation breaks down as in Sec. II, in which case the magnetic moment μ is obviously not conserved. In fact, it was shown via test-particle simulations[32] that regions where μ is not conserved coincide with regions where the stochastic ion heating condition is satisfied, and that initially close particles diverge in time—i.e., the system has a positive Lyapunov exponent.

3. *Temperature anisotropy.* Stochastic heating is associated with perpendicular motion and thus with perpendicular temperature. The perpendicular energy gain, however, could eventually be redistributed in the parallel direction through other means such as collisions, waves, and/or instabilities.

4. *Reduction of the heating amount and extent with an increasing guide field.* This correlation was predicted in Sec. IV and verified in Sec. V. In addition, the analysis in Sec. III also hints to this correlation. For instance, including a guide field B_{z0} when inserting Eq. 37 into Eq. 1 yields the condition,
$$L_x^2 < d_i^2 \frac{B_y^2}{B_y^2 + B_{z0}^2},$$
which is a stronger requirement than Eq. 38 for finite B_{z0}.

5. *Preferential heating of ions with higher mass-to-charge ratios.* The stochastic ion heating condition (Eq. 1) is more easily satisfied for higher values of m_i/q_i. The heating amount also increases, as evidenced by PIC simulations with a partial mixture of heavy ions in Yoon and Bellan[32].

It should be mentioned that there may of course be other reconnection situations where the dominant ion heating mechanism is not stochastic heating. These situations may for example involve three-dimensional structures, high plasma beta, and/or high guide field strengths; as mentioned in Sec. I, other mechanisms could then cause the ion heating. Also, there may also be situations where ion heating is due to stochastic heating but the source for large α differs (e.g., lower-hybrid drift waves[43,44,51] in asymmetric reconnection[52,53]). However, for the textbook two-dimensional collisionless reconnection geometry which has been the focus of numerous previous studies, stochastic ion heating is not only important but intrinsic to the process. Therefore, the attribution of ion heating to the stochastic heating mechanism due to the Hall electric fields must first be considered when analyzing any reconnection process.

VII. CONCLUSION

It has been shown from various perspectives that stochastic heating is the anomalous ion heating mech-

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anism that is intrinsic to collisionless magnetic reconnection. The in-plane Hall electric fields that intrinsically develop during collisionless reconnection both energize and chaotize ions, effectively heating the species. The mechanism is consistent with the main observed features of ion heating during reconnection.

ACKNOWLEDGMENTS

The authors would like to thank Gunsu S. Yun for useful discussions and suggestions. This material is based upon work supported by the U.S. Department of Energy Office of Science, Office of Fusion Energy Sciences under award No. DE-FG02-04ER54755, the Air Force Office of Scientific Research under award No. FA9550-17-1-0023, and the NSF Solar Terrestrial Research Program via award grant 1914599. The particle-in-cell computations presented here were conducted on the Caltech High Performance Cluster partially supported by a grant from the Gordon and Betty Moore Foundation.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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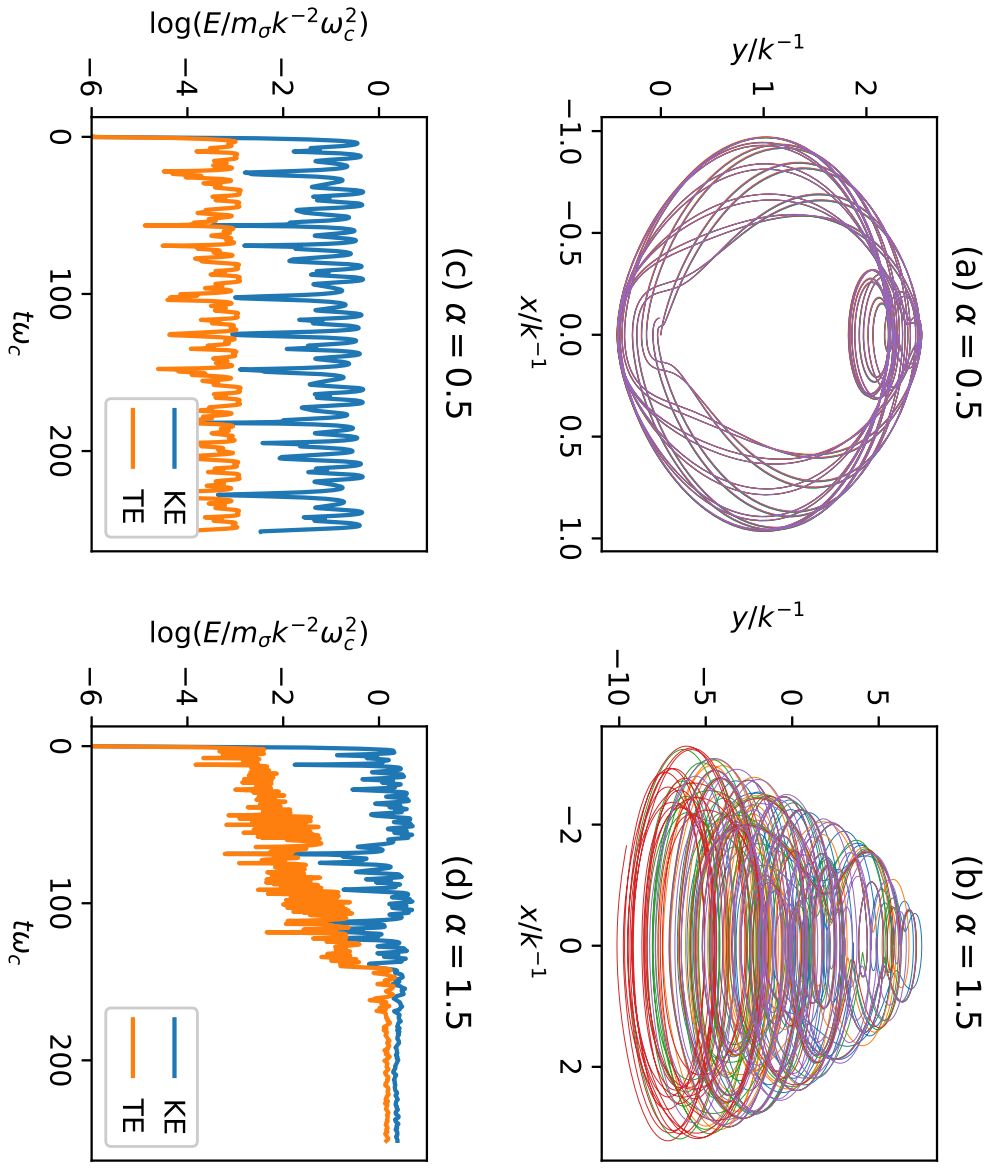
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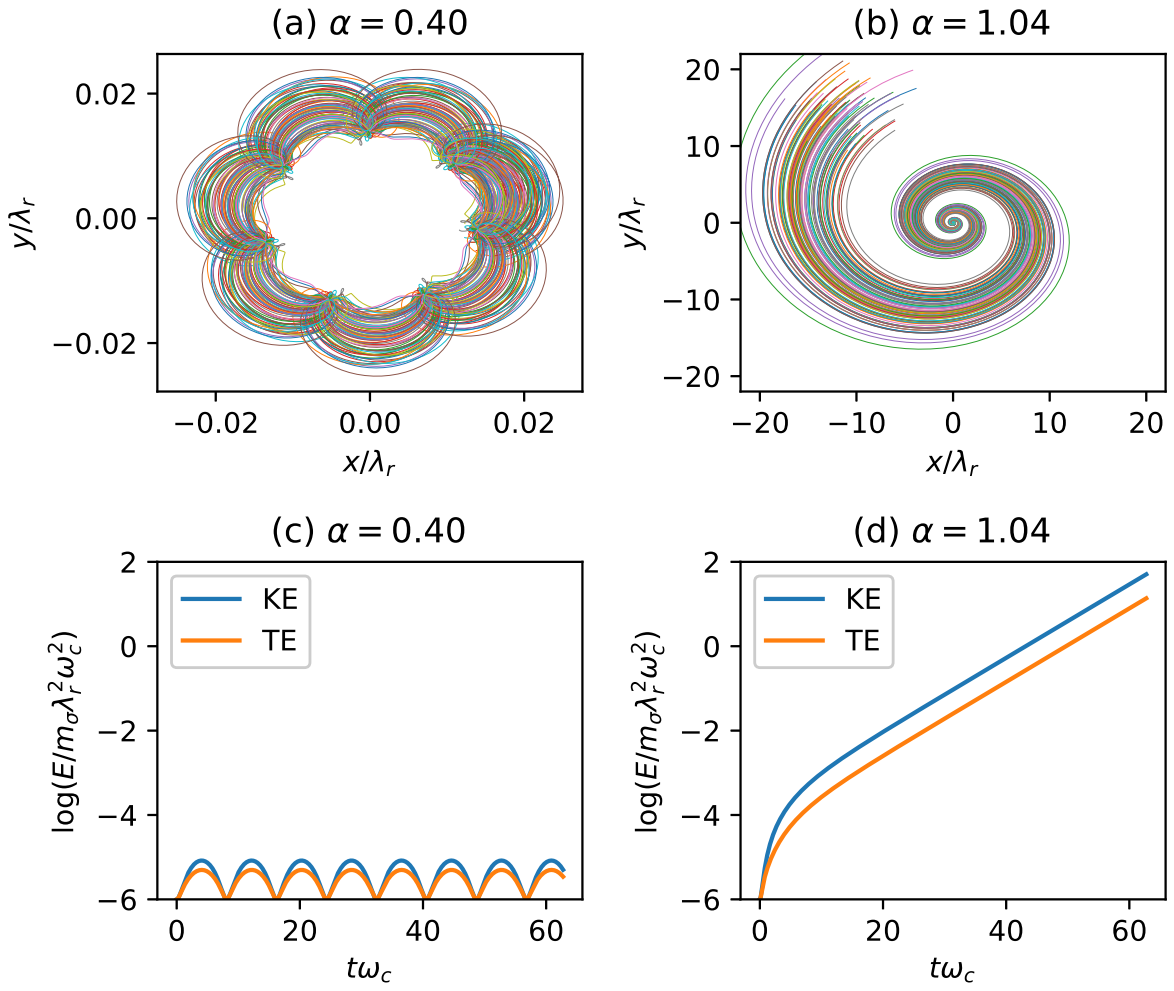
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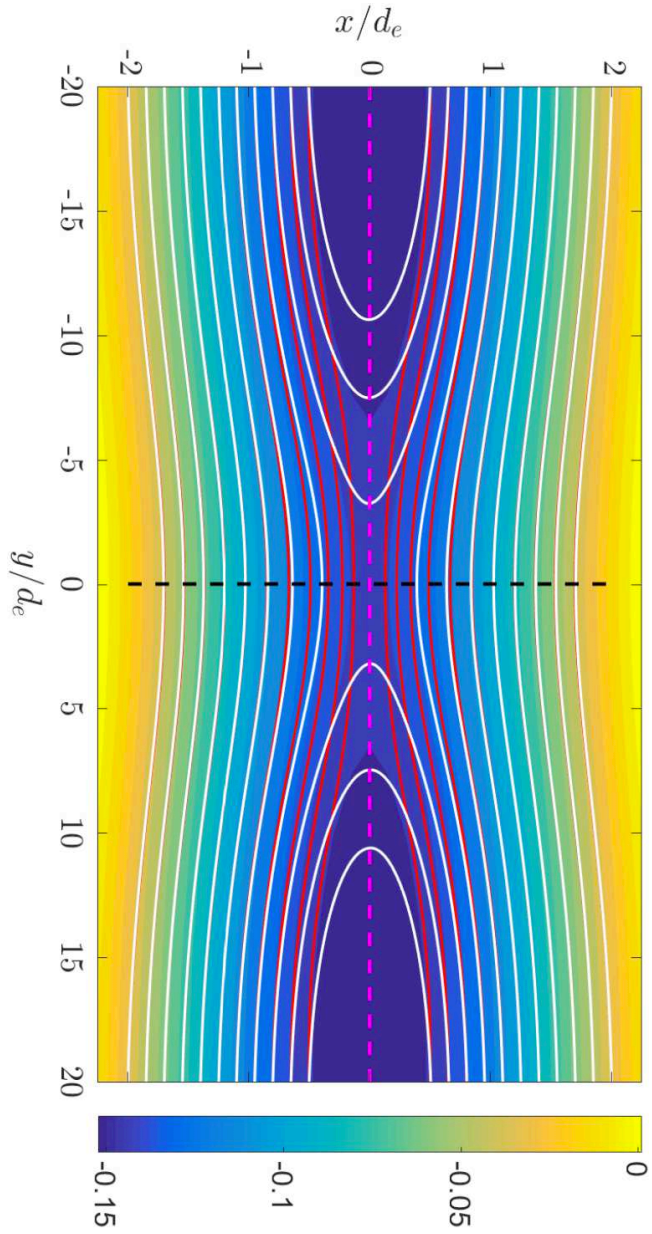
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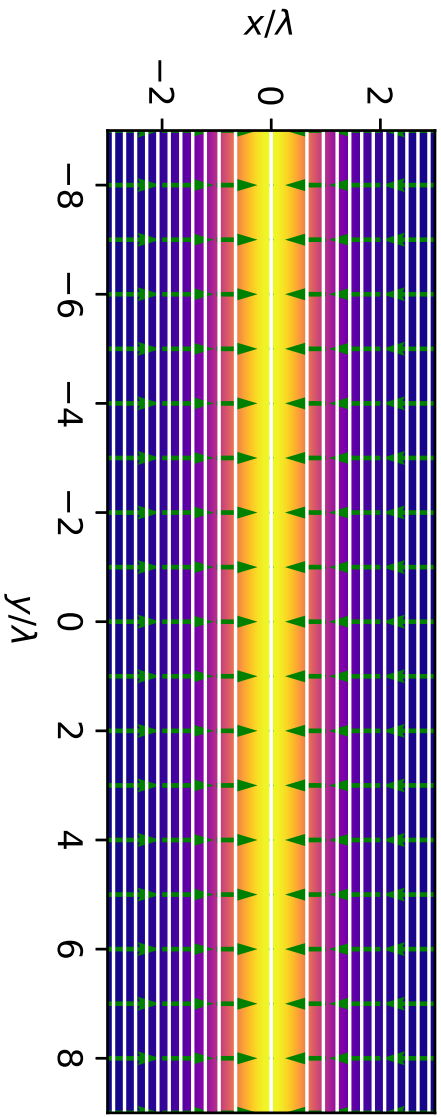
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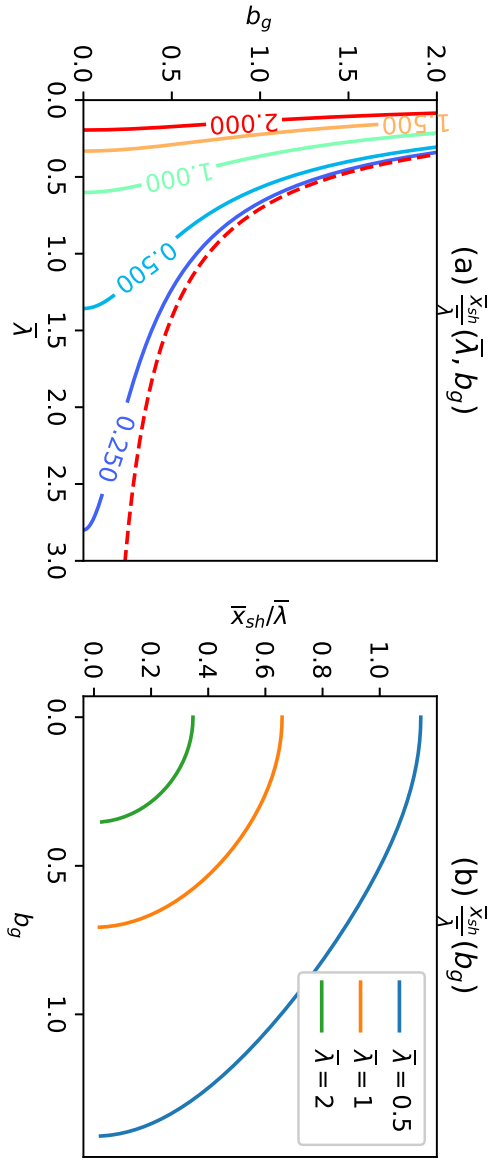
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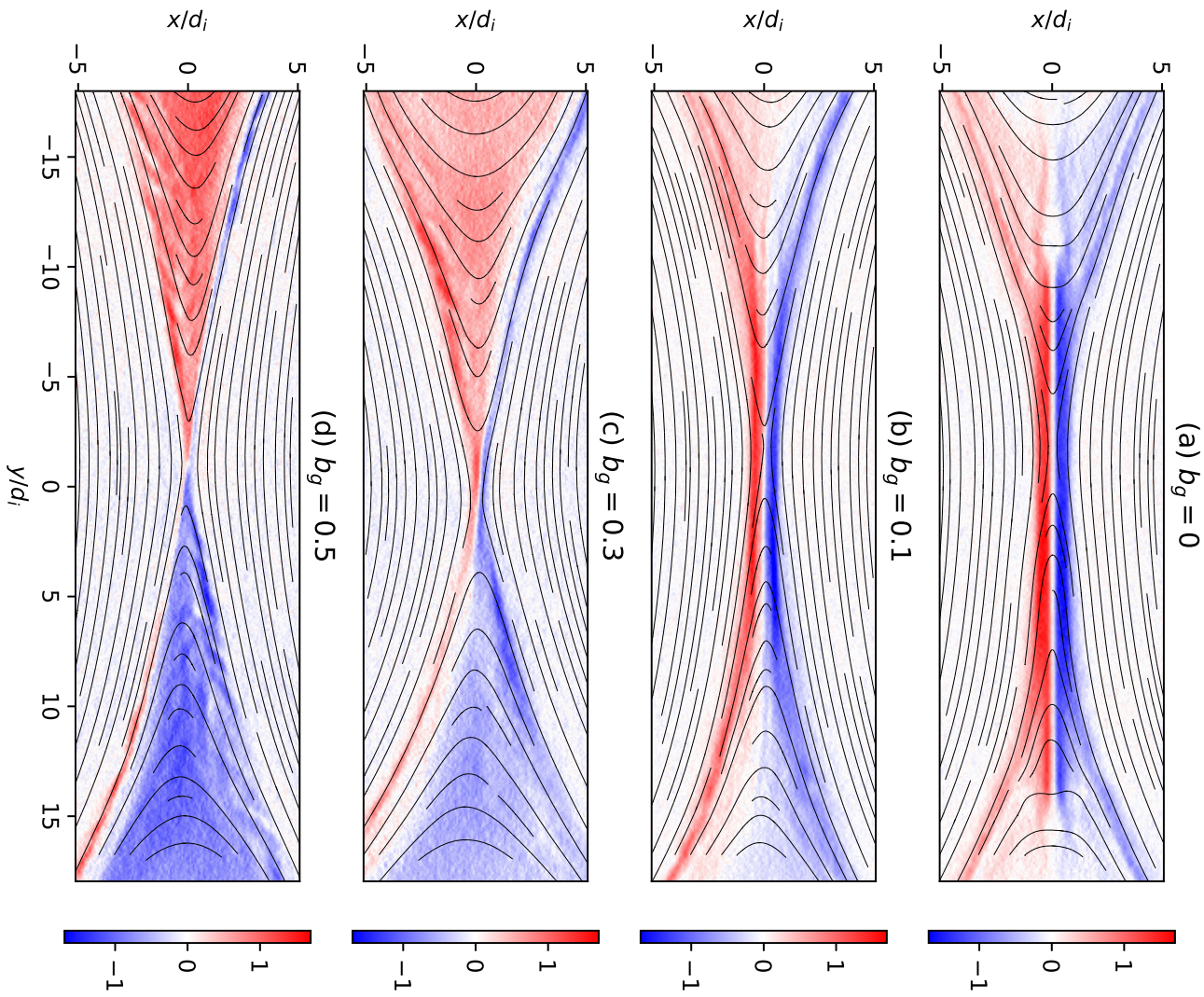
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